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Prop fn exists var aleatoria

Proposición 1. *Sea* $F\colon \mathbb{R} \longrightarrow \mathbb{R}$ *monótona no decreciente que cumple* (1), (2) *y* (3)

$$\Longrightarrow \exists\,(\Omega, \Sigma, \mathbb{P}) \text{ espacio de probabilidad}\colon \exists X \text{ v.a.} : \forall t \in \mathbb{R} : F_X(t) = F(t).$$

***Demostración*:** Consideramos en el espacio de probabilidad $([0,1], \mathcal{B}([0,1]), m)$ la variable aleatoria $X\colon [0,1] \longrightarrow \mathbb{R}$ dada por $\forall \omega \in [0,1] : X(\omega) = \sup\{t \in \mathbb{R} : F(t) < \omega\}$.

$$\begin{aligned}\Longrightarrow \forall t \in \mathbb{R} : m(\{\omega \in \mathbb{R} : X(\omega) \leqslant t\}) &= m(\{\omega \in \mathbb{R} : \sup\{s \in \mathbb{R} : F(s) < \omega\} \leqslant t\}) \\ &= m(\{\omega \in \mathbb{R} : \omega \leqslant F(t)\}) = F(t).\end{aligned}$$

Por tanto, $\forall t \in \mathbb{R} : F_X(t) = m(\{\omega \in \mathbb{R} : X(\omega) \leqslant t\}) = F(t)$[1]. ■

[1] También se puede definiendo X como la identidad en el espacio de medida dado por la medida de Lebesgue-Stieltjes.

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